A16: the folded table's scan arrays collapsed onto 2 MiB pages

Exploration, no round. One binary (`ic-a16-v8-e7136eb8`, `huge-pages.patch` on v8) with three arms set by `KIC_HUGE_PAGES`:

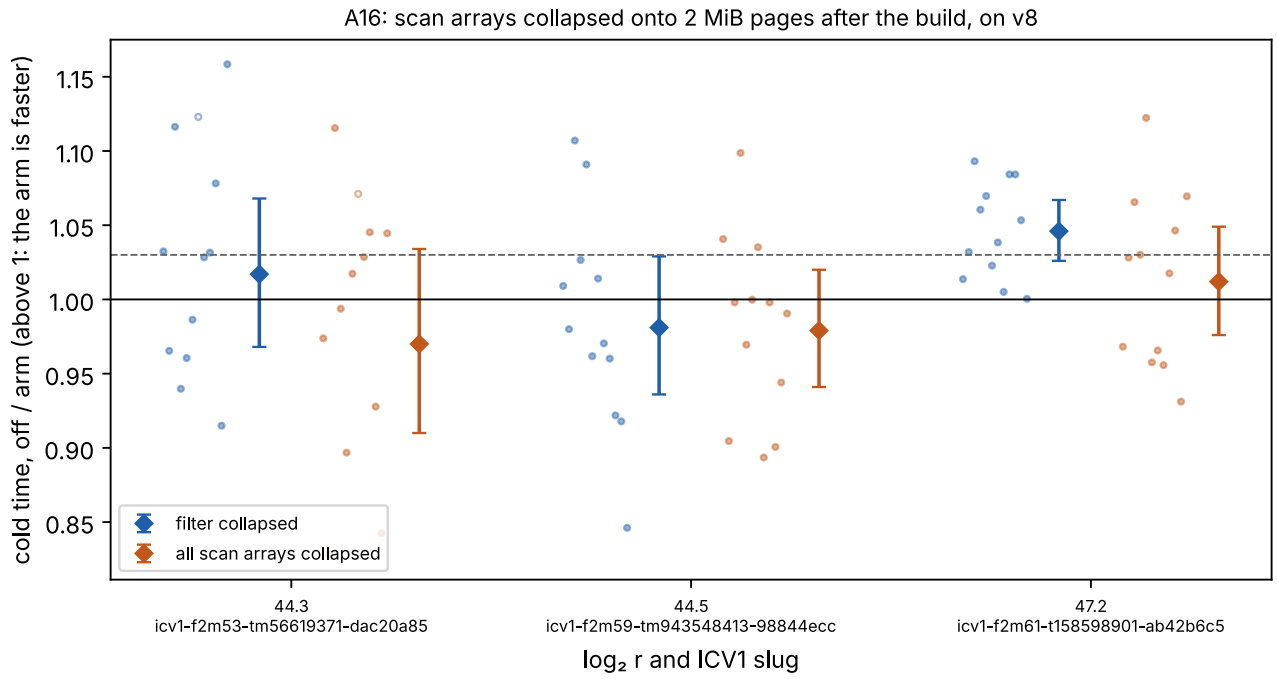
- `off`: v8's pages;
- `filter`: the filter alone collapsed;
- `all`: the filter, the bucket starts and the stored keys collapsed.

Collapsing is `MADV_HUGEPAGE`, then `MADV_COLLAPSE`, after the build. The runs: 2026-10-17 04:24–04:54 UTC on the Emerald Rapids host R12 ran on; `M1`'s two rows at the three target sizes; six rounds with the order rotating; 108 isolated processes, all completed.

- **Contended runs:** one (`off`, 2^44.3, round 1), excluded.
- **Logarithms:** every pair recovered the same logarithm in all three arms.

Outcome:

- **The filter alone** gains 4.6% cold at 2^47.2 ([1.026, 1.067]) and is inconclusive at 2^44.3 and 2^44.5.
- **Every array** gains nothing: the build pays for collapsing the stored keys.
- **Why the filter's gain is partial:** only the whole 2 MiB pages inside the filter's block collapse, and the block starts off a boundary.



explorations/A16-huge-pages-20261017/pairs-*.tsv; 108 isolated processes; open: a contended run; dashed: 1.03

Figure 1: Cold time, `off` over each arm, per pair (open: a contended run), with the clean pairs' geometric mean and 95% t interval. Dashed: 1.03.

curve	$\log_2 r$	off / filter [95%]	off / all [95%]	collection off / filter
`icv1-f2m53-tm56619371-dac20a85`	44.3	1.017 [0.968, 1.068]	0.970 [0.910, 1.034]	1.027
`icv1-f2m59-tm943548413-98844ecc`	44.5	0.981 [0.936, 1.029]	0.979 [0.941, 1.020]	1.018
`icv1-f2m61-t158598901-ab42b6c5`	47.2	1.046 [1.026, 1.067]	1.012 [0.976, 1.049]	1.062

Smoke records (single processes at 2^47.2, before the timed runs; observations, not measurements):

- glibc's `malloc.hugetlb=1` on every allocation made collection faster (1395 ⟶ 1177 ms) but the build slower (213 ⟶ 565 ms).
- A16's full collapse gave cold 1626 ⟶ 1558 ms.

Scope: one host, one core, three sizes, two rows a size. THP is `madvise` here.